DOUBLY ISOLATED BATYREV MIRROR PAIRS AND NON-SMOOTHABLE CALABI–YAU THREEFOLDS

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Abstract. For a reflexive 4-polytope Δ with unit edges the generic anticanonical hypersurface $X \subset \mathbb{P}_\Delta$ is a Calabi–Yau threefold with isolated singularities; we ask when the Batyrev mirror $X^\circ \subset \mathbb{P}_{\Delta^\circ}$ is isolated-singular as well. An exact identity — the lattice length of the dual edge of a two-dimensional face equals the multiplicity of its transverse cone — shows this happens if and only if every two-dimensional face of Δ is transversally smooth; we call such Δ *both-sides unit*. We determine the both-sides unit polytopes completely: among the 473,800,776 reflexive 4-polytopes of the Kreuzer–Skarke classification there are exactly 590, and every two-dimensional face of every one of them is a triangle, a zonotope, or a reflexive polygon (one interior point). Consequently the non-smoothable germs occurring on doubly isolated mirror pairs are exactly the cyclic quotient points and the anticanonical cone over $\mathbb{F}_1$ — and the deformable-but-non-smoothable germs of the first paper of this series never occur, so an isolated-singular Calabi–Yau threefold carrying one never has an isolated-singular mirror. At the opposite extreme, exactly one reflexive 4-polytope — the self-dual 24-cell, whose hypersurface is the self-mirror threefold with Hodge numbers $(20, 20)$ — yields a mirror pair with both members *smooth*. The $\mathbb{F}_1$-cone occurs exactly once: a unique pair of polytopes, with 22 and 26 vertices, whose hypersurfaces form a mirror pair with resolution Hodge numbers $(20, 26)$ and $(26, 20)$ in which X is non-smoothable while every singular point of X° is locally smoothable. We also record, as a caution, that the natural conjecture suggested by the low-vertex data — that only triangles and zonotopes occur — is true for all 395,406,329 polytopes with at most 18 vertices and false from 19 vertices on.

1. Introduction

Let $\Delta \subset N_\mathbb{R} = \mathbb{R}^4$ be a reflexive polytope all of whose edges have lattice length one, let $\Delta^\circ \subset M_\mathbb{R}$ be its polar dual, and let $X \in |-K_{\mathbb{P}_\Delta}|$ be a generic anticanonical hypersurface in the Gorenstein Fano toric fourfold associated to the face fan of Δ. By Batyrev [2], X is a Calabi–Yau threefold with Gorenstein canonical singularities, and by the planting proposition of [8, Prop. 3.1] its singular locus is exactly: for each two-dimensional face $F \preceq \Delta$ whose cone $C(F)$ is singular, $\ell(F^\circ)$ points at which the germ of X is analytically isomorphic to $C(F)$, where $\ell(F^\circ)$ is the lattice length of the edge $F^\circ \preceq \Delta^\circ$ dual to F. The trichotomy of [8] classifies $C(F)$ by the combinatorics of F into rigid (R), deformable-but-non-smoothable (D), and smoothable (S); a single face of the first two types forces X to admit no smoothing, and [9] studies when the type-(S) configurations smooth.

Mirror symmetry, in the Batyrev picture, is the polar duality $\Delta \leftrightarrow \Delta^\circ$, which for 4-polytopes exchanges two-dimensional faces with edges. This note asks the mirror question: when are X and the mirror $X^\circ \subset \mathbb{P}_{\Delta^\circ}$ *both* isolated-singular, and which local models — in particular, which non-smoothable ones — can occur in that doubly isolated situation?

The entry point is exact and elementary. For a two-dimensional face F write $u_1, u_2 \in M$ for the primitive inner normals of the two facets of Δ containing F, so $F^\circ = \mathrm{conv}\{u_1, u_2\}$, and let $M_F = M \cap (\mathbb{R}u_1 + \mathbb{R}u_2)$.

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Theorem 1.1 (= Lemma 3.1 with Remark 3.2)**.** *$\ell(F^\circ) = [\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2 \,]$, the multiplicity of the normal cone* Cone(u_1, u_2) *of F — equivalently, of its dual, the transverse cone of Δ along F. In particular $\ell(F^\circ) = 1$ if and only if the transverse cone is smooth.*

Since X is isolated-singular exactly when Δ has unit edges, and X° exactly when every two-dimensional face of Δ has $\ell(F^\circ) = 1$, we obtain:

Corollary 1.2. *X and X° are both isolated-singular if and only if Δ has unit edges and every two-dimensional face of Δ is transversally smooth. We call such Δ* both-sides unit*. The condition is self-dual: Δ is both-sides unit if and only if Δ° is.*

Our main results determine the both-sides unit polytopes, and the local models they carry, *completely*. They rest on an exhaustive scan of the Kreuzer–Skarke classification [7] of all 473,800,776 reflexive 4-polytopes, whose methodology — including the guards against false negatives — is described in §5; every polytope found is re-verified in exact arithmetic, so the statements below are theorems with computer-assisted (finite, exhaustive) proofs.

Theorem 1.3 (classification of both-sides local models)**.** *There are exactly* 590 *both-sides unit reflexive* 4*-polytopes. Every two-dimensional face of every one of them is*

- *a triangle (the germ is a cyclic quotient point, or smooth), or*
- *a zonotope — a centrally-symmetric unit-edge polygon, always of type* (S) *— or*
- *a reflexive polygon: one of the five unit-edge polygons with exactly one interior lattice point, i.e. the anticanonical polygons of the five smooth toric del Pezzo surfaces.*

All five reflexive polygons occur. The complete face inventory is given in Table 2.

Theorem 1.4 (the non-smoothable germs of doubly isolated pairs)**.** *Exactly three of the* 590 *both-sides unit polytopes carry a non-smoothable two-dimensional face, and the germs occurring are: cyclic quotient triangles —* $[3, 1] = \frac{1}{3}(1, 1, 1)$*, six times on one* 9*-vertex polytope, and* $[3, 2]$*, ten times on one* 5*-vertex simplex — and the anticanonical cone over $\mathbb{F}_1$, exactly* once*, on a unique* 22*-vertex polytope. In particular no deformable-but-non-smoothable* (D) *face occurs on any both-sides unit polytope.*

Corollary 1.5. *Let X be a unit-edge Batyrev Calabi–Yau threefold with a singular point of type* (D) *— for instance the pentagon-cone threefolds of* [8, Thms. 5.3, 5.7]*. Then the Batyrev mirror X° is non-isolated: it carries a curve of transverse A_{m-1} singularities for some $m \geq 2$. The deformable-but-non-smoothable phenomenon never occurs on a doubly isolated mirror pair.*

The one non-quotient exception is the most interesting object of this note.

Theorem 1.6 (the unique frozen mirror pair)**.** *Up to lattice isomorphism there is exactly one doubly isolated Batyrev mirror pair whose singularities are not all quotient points or smoothable: the pair $(\Delta_{\mathbb{F}_1}, \Delta^\circ_{\mathbb{F}_1})$ of Theorem 6.1, with* 22 *and* 26 *vertices. The generic anticanonical $X \subset \mathbb{P}_{\Delta_{\mathbb{F}_1}}$ has* 20 *ordinary double points, one* dP_6*-cone point, and one point whose germ is the anticanonical cone over $\mathbb{F}_1$; hence X* admits no smoothing [8, Cor. 4.3]*. Its mirror X° has* 26 *ordinary double points, two* dP_7*-cone points and two* dP_6*-cone points — every singular germ of X° is of type* (S)*. The maximal projective crepant partial resolutions have* $(h^{1,1}, h^{2,1}) = (20, 26)$ *and* $(26, 20)$ *respectively.*

Thus the Batyrev mirror of a Calabi–Yau threefold that is *frozen* — non-smoothable, with the very germ excluded in Gross's smoothing theorem for primitive contractions [6, Thm. 5.8] — can be isolated-singular with every singular point locally smoothable. Notably, every dual edge on the mirror side has length one, so X° lies entirely in the $\ell(F^\circ) = 1$ regime of [9]: its 26 nodes pose the cross-face Friedman problem of [9, §7], and its del Pezzo cone points are the germs that [9, §6] places beyond Gross's theorems and for which no Friedman-type obstruction is known; whether X° is globally smoothable is open (Question 7.2).

Theorems 1.3, 1.4 and 1.6 answer the mirror question of [8, §6.3] for the non-smoothable stratum: among the non-smoothable local models, only the cyclic quotients and the $\mathbb{F}_1$-cone — precisely the germs of Gross's two exceptional surfaces $\mathbb{P}^2$ and $\mathbb{F}_1$ [6, Thm. 5.8] — admit doubly isolated mirror realizations, and the $\mathbb{F}_1$-cone does so exactly once in the entire toric hypersurface landscape.

The opposite extreme of the classification carries a matching uniqueness. Exactly eight of the 590 polytopes have all two-dimensional faces smooth, so that X itself is smooth; for exactly one of them — the self-dual 24-cell, whose hypersurface is the classical self-mirror threefold with $(h^{1,1}, h^{2,1}) = (20, 20)$ [3] — the mirror is smooth as well. The smooth Batyrev mirror pair is thus unique in the entire classification (Example 5.4).

Two refuted conjectures, and a caution about low-vertex evidence. This note originally conjectured that every two-dimensional face of a both-sides unit polytope is a triangle or a zonotope — equivalently, that no *asymmetric* face, one that is neither a triangle nor centrally symmetric, ever occurs — and reduced this to a local "balancing" statement about a face and its star. Both statements are *false*: the dP_7 pentagon occurs (87 faces, on polytopes with 19 to 31 vertices), as does the $\mathbb{F}_1$ quadrilateral (once). What makes the failure instructive is its location. The conjecture is *true* for all 395,406,329 reflexive 4-polytopes with at most 18 vertices — 83.5% of the classification — and fails from 19 vertices on, where both-sides unit polytopes first become common (Table 1); the local statement had itself survived a complete check of the 38,571 candidate configurations arising among the 1,037,834 polytopes with at most eight vertices. Small-vertex evidence, however extensive, samples a biased corner of the landscape; we record the episode as a data point on the epistemology of conjecture-making over the Kreuzer–Skarke list.

All computations — the identity, the scan, the exact re-verification of all 590 polytopes, and the mirror pair — are performed by exact-arithmetic programs building on those of [8], included as ancillary files; `both_sides_census.py` re-derives every number and theorem in this note from the committed scan results, with assertions.

Acknowledgements. This note completes the series [8, 9]; the question of smoothing canonical Calabi–Yau singularities was posed to the author by Mark Gross many years ago.

2. Preliminaries

We keep the notation of [8]. For a lattice polygon Q with unit edges placed at height one, $C(Q)$ denotes the associated Gorenstein toric affine threefold; Q is of type (R), (D) or (S) according as its edge-vector multiset has no proper zero-sum sub-multiset, has one but no partition into unit segments and unimodular triples, or has such a partition [1, 4]. A unit-edge polygon is a *zonotope* — a Minkowski sum of unit segments — if and only if it is centrally symmetric, and every zonotope is of type (S). A unit-edge polygon with exactly one interior lattice point is reflexive; there are five, the anticanonical polygons of $\mathbb{P}^2$, $\mathbb{F}_1$, $\mathbb{P}^1\times\mathbb{P}^1$, dP_7, dP_6, of which the $\mathbb{P}^2$ triangle and the $\mathbb{F}_1$ quadrilateral are rigid and the other three are of type (S) [8, Rem. 2.6]. For a reflexive 4-polytope Δ we write $\Delta^\circ = \{u : \langle u, v\rangle \geq -1\ \forall v \in \Delta\}$; a face $G \preceq \Delta$ has dual face $G^\circ \preceq \Delta^\circ$ with $\dim G + \dim G^\circ = 3$. By the planting proposition [8, Prop. 3.1], for Δ with unit edges the generic $X \in |-K|$ has, over each two-dimensional face F with $C(F)$ singular, exactly $\ell(F^\circ)$ singular points with germ $C(F)$, and no other singular points; and an edge of Δ of lattice length $m \geq 2$ produces a curve of transverse A_{m-1} singularities [8, Rem. 3.2].

3. The transverse-multiplicity identity

Let F be a two-dimensional face of a reflexive polytope $\Delta \subset N_{\mathbb{R}}$, and let $u_1, u_2 \in M$ be the primitive inner normals of the two facets of Δ containing F, so that $\langle u_i, \cdot\rangle = -1$ on F and $F^\circ = \operatorname{conv}\{u_1, u_2\}$. (The u_i are vertices of Δ°; the facet tables of [8] list the sign-flipped functionals taking the value $+1$ on the facet.) Put $M_F = M \cap (\mathbb{R}u_1 + \mathbb{R}u_2)$, a rank-two saturated sublattice of M containing u_1, u_2 as primitive vectors.

Lemma 3.1. $\ell(F^\circ) = [\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2\,]$.

Proof. Both sides depend only on the pair (u_1, u_2) inside the rank-two lattice M_F. Choose a $\mathbb{Z}$-basis of M_F in which $u_1 = (0, -1)$; this is possible because u_1 is primitive. Write $u_2 = (p, q)$; then $p = \det(u_1, u_2) = \pm[\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2\,]$, and $\ell(F^\circ) = \ell\big(\operatorname{conv}\{u_1, u_2\}\big) = \gcd(u_2 - u_1) = \gcd(p,\, q + 1)$.

Let v be a vertex of F and let $\bar{v} \in N_F := N/(N \cap u_1^\perp \cap u_2^\perp)$ be its image; since u_1, u_2 vanish on $N \cap u_1^\perp \cap u_2^\perp$ they descend to elements of $M_F = N_F^\vee$, and $\bar{v}$ is a lattice point of $N_F \cong \mathbb{Z}^2$ with $\langle u_i, \bar{v}\rangle = -1$. Writing $\bar{v} = (a, b)$ in the basis of N_F dual to the chosen basis of M_F, $\langle u_1, \bar{v}\rangle = -b = -1$ gives $b = 1$, and $\langle u_2, \bar{v}\rangle = pa + q = -1$ gives $pa = -1 - q$, so $p \mid (q + 1)$. Hence $\gcd(p, q + 1) = |p|$, and $\ell(F^\circ) = |p| = [\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2\,]$. □

Remark 3.2. Lemma 3.1 is elementary and may be known; what matters here is the reading it provides. The right-hand side is the multiplicity of the two-dimensional cone $\operatorname{Cone}(u_1, u_2)$, which is the normal cone of F and dually the transverse cone of Δ along F (for two-dimensional cones the multiplicities of a cone and its dual agree). Thus *the dual-edge length counts exactly the transverse singularity multiplicity*, and $\ell(F^\circ) = 1$ is the statement that Δ is transversally smooth along F. (Transversal smoothness concerns only the directions normal to F: the germ $C(F)$ itself may well be singular.) The lattice-point hypothesis on v is essential: for an arbitrary pair of primitive vectors the two sides can differ.

Corollary 1.2 is immediate: X is isolated-singular iff Δ has unit edges; X° is isolated-singular iff Δ° has unit edges, i.e. every edge F° of Δ° has $\ell(F^\circ) = 1$, i.e. by Lemma 3.1 every two-dimensional face of Δ is transversally smooth. Self-duality follows since the conditions "unit edges" and "all dual edges of length one" are exchanged by $\Delta \leftrightarrow \Delta^\circ$.

4. The mirror correspondence of the singular strata

Proposition 4.1. *Let Δ be reflexive with unit edges and F a two-dimensional face with $C(F)$ singular. Under polar duality F corresponds to the edge F° of Δ°, and:*

1. *X has $\ell(F^\circ)$ isolated points of type $C(F)$ over F;*
2. *X° has, along the surface $V(\sigma_{F^\circ}) \subset \mathbb{P}_{\Delta^\circ}$ — the orbit closure of the cone σ_{F° over the edge F° in the face fan of Δ° — a curve of transverse $A_{\ell(F^\circ)-1}$ singularities (and none if $\ell(F^\circ) = 1$).*

In particular X° is isolated-singular if and only if $\max_F \ell(F^\circ) = 1$, i.e. Δ is both-sides unit.

Proof. (1) is [8, Prop. 3.1]. For (2), the edge F° of Δ° has lattice length $\ell(F^\circ)$; an edge of lattice length m produces on the generic anticanonical hypersurface a curve of transverse A_{m-1} singularities (§2), and none when $m = 1$. Hence X° is isolated-singular iff every two-dimensional face of Δ has $\ell(F^\circ) = 1$. □

Thus the $\ell(F^\circ)$-point packet of X over F and the $A_{\ell(F^\circ)-1}$-curve of X° are mirror-dual, and the two threefolds are simultaneously isolated-singular exactly on the both-sides unit polytopes. Whenever Δ has *any* face with $\ell(F^\circ) \geq 2$ — the generic situation — the mirror is non-isolated and its singularities leave the isolated framework of [1, 4], being governed instead by Filip's non-isolated 0-mutability criterion [5].

5. The census of both-sides unit polytopes

5.1. **Methodology.** Both-sides unitality is a face-level test: all edges of lattice length one and, by Lemma 3.1, all dual edges of length one. We ran it over the complete Kreuzer–Skarke classification with the exact-arithmetic integer engine built for the scan of [8, §6.1]. The fast test computes the defining predicate itself in exact integer arithmetic — no floating point, no heuristic pruning — so exhaustiveness reduces to implementation correctness, which is controlled in three independent ways: before scanning each database file the engine was re-checked against a separately written reference implementation on random rows of that file; the 3,905,789 polytopes with at most nine vertices were scanned in full by both implementations, with identical results; and the 590 polytopes found pass a global polarity audit — the both-sides condition is self-dual (Corollary 1.2), and for every hit the polar, computed exactly, is again both-sides unit and matches a hit in vertex count, facet count and complete face inventory, an invariance that a false negative whose polar partner was detected would violate. The one polytope missing from the per-vertex-count mirror of the database — the 36-vertex hexagon product, see [8, §6.1] — is included. Every polytope passing the fast test is then re-verified, and its faces classified, by the exact reference toolkit; the chain driver among the ancillary files reproduces the full scan from the public database. The scan results are committed as data files, and the ancillary program `both_sides_census.py` re-derives every statement of Theorems 1.3, 1.4 and 1.6 from them, with assertions.

vertices	5	6	8	9	10	12	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	33	36
#	2	1	1	1	1	1	1	4	4	7	12	24	37	69	84	98	85	64	42	24	13	9	4	1	1

Table 1. The 590 both-sides unit reflexive 4-polytopes by vertex count (no others occur; in particular none with 7, 11, 13, 14, 32, 34 or 35 vertices). The population is closed under polar duality (Corollary 1.2) and peaks at 24 vertices.

5.2. **The classification.**

Proof of Theorems 1.3 and 1.4. Exhaustive verification, summarized in Tables 1 and 2. The scan of all 473,800,776 polytopes returned 590 both-sides unit ones; exact re-analysis of each classifies all their two-dimensional faces into the twelve classes of Table 2, each of which is a triangle, a zonotope, or one of the five reflexive polygons (the $[4,1]$ zonotope is the $\mathbb{P}^1\times\mathbb{P}^1$ diamond, the $[6,1]$ zonotope the dP_6 hexagon, the $[3,1]$ triangle the $\mathbb{P}^2$ triangle; the asymmetric $[5,1]$ and $[4,1]$ classes are the dP_7 pentagon and the $\mathbb{F}_1$ quadrilateral, identified by explicit $GL_2(\mathbb{Z})$-equivalences). The non-smoothable classes occurring are the rigid triangles $[3,1]$, $[3,2]$ and the $\mathbb{F}_1$ quadrilateral; the $[3,2]$ faces all lie on the 5-vertex quotient simplex of Example 5.2, the $[3,1]$ faces on the 9-vertex polytope of Example 5.2, and the $\mathbb{F}_1$ face on the unique polytope of Theorem 6.1. No face of type (D) occurs. ∎

face $[k,i]$	kind	count	germ of X over it
$[3,0]$	triangle	25,873	smooth point
$[3,1]$	triangle	6	$\frac{1}{3}(1,1,1)$ **(rigid)**
$[3,2]$	triangle	10	cyclic quotient **(rigid)**
$[4,0]$	zonotope	15,652	ordinary double point
$[4,1]$	zonotope	99	cone over $\mathbb{P}^1 \times \mathbb{P}^1$
$[4,1]$	*asymmetric*	1	cone over $\mathbb{F}_1$ **(rigid)**
$[4,2]$	zonotope	9	type (S)
$[5,1]$	*asymmetric*	87	cone over dP_7
$[6,1]$	zonotope	395	cone over dP_6, 2 smoothing components
$[6,2]$	zonotope	2	type (S)
$[6,7]$	zonotope	3	type (S)
$[8,4]$	zonotope	1	type (S), 3 smoothing components

TABLE 2. Every two-dimensional face of every both-sides unit polytope, by class (k = edges, i = interior points). "Asymmetric" = neither a triangle nor centrally symmetric; only the two non-centrally-symmetric reflexive polygons occur, and each germ appears on X exactly once per face since all dual edges have length one. Type-(S) germs have one smoothing component except where noted.

Proof of Corollary 1.5. A type-(D) point of X is the cone over a type-(D) face F. By Theorem 1.4, Δ is then not both-sides unit, so some face F' has $\ell(F'^{\circ}) \geq 2$ and Proposition 4.1(2) gives a singular curve on X°. □

5.3. **The refuted conjectures.** The first version of this note conjectured that every two-dimensional face of a both-sides unit polytope is a triangle or a zonotope, and reduced this to a local claim: an asymmetric face with unit-edge star and $\ell(F^{\circ}) = 1$ forces a star face with $\ell \geq 2$. Table 2 refutes both: 88 asymmetric faces occur, on 55 polytopes. The smallest counterexample has 19 vertices:

$$\begin{gathered}\Delta_{19} = \operatorname{conv}\{(1,0,0,0),\ (0,1,0,0),\ (0,0,1,0),\ (0,0,-1,0),\ (0,-1,0,0),\\ (1,-1,-1,0),\ (0,0,0,1),\ (-1,1,1,-1),\ (0,-1,0,1),\ (0,0,-1,1),\\ (-1,1,0,-1),\ (-1,0,1,-1),\ (-1,0,1,0),\ (-1,1,0,0),\ (0,-1,-1,0),\\ (0,-1,-1,1),\ (-1,0,0,-1),\ (-2,1,1,-1),\ (-1,0,0,1)\},\end{gathered}$$

a both-sides unit polytope one of whose faces is the dP_7 pentagon; its hypersurface is a Calabi–Yau threefold with 14 nodes and one dP_7-cone point, all with isolated-singular mirror. The conjecture is *true* on the 395,406,329 polytopes with at most 18 vertices and false from 19 on — precisely where both-sides unit polytopes stop being sporadic (Table 1). We let the corrected statement stand as Theorem 1.3, and the conceptual question it raises as Question 7.1.

Remark 5.1 (why not "simplicial"). The stronger guess that a both-sides unit polytope is simplicial fails for a structural reason: the condition is self-dual, and "simplicial and simple" forces a simplex, contradicting the existence of both-sides unit polytopes with more than five vertices. The non-triangular faces are genuine; the content of Theorem 1.3 is that they are zonotopes or reflexive.

Example 5.2 (the low-vertex stratum)**.** Among the 3,905,789 polytopes with at most nine vertices exactly five are both-sides unit; four of them form two polar pairs. One pair is the smooth simplex and the simplex

$$\operatorname{conv}\{(1,0,0,0),\ (0,1,0,0),\ (2,4,5,0),\ (3,3,0,5),\ (-6,-8,-5,-5)\},$$

whose ten faces are the $[3,2]$ rigid triangles of Table 2 (cyclic quotient points) and whose polar is smooth. A second pair consists of a simplicial six-vertex polytope with all faces smooth and its nine-vertex polar, carrying the six $\frac{1}{3}(1,1,1)$ triangles and nine $[4,2]$ zonotopes. In each pair, the non-smoothable member mirrors a *smooth* threefold. The fifth polytope — eight vertices, sixteen facets, all thirty-two faces smooth triangles, so its hypersurface X is smooth — shows that the polarity closure of Corollary 1.2 does not respect vertex-count windows: its polar is a sixteen-vertex both-sides unit polytope all twenty-four of whose faces are $[4,1]$ diamonds, so the mirror X° carries twenty-four cone-over-$\mathbb{P}^1\times\mathbb{P}^1$ points.

Example 5.3 (the top of the classification)**.** The 36-vertex product $H \times H$ of two reflexive hexagons — the reflexive 4-polytope with the most vertices, and the one polytope absent from the per-vertex-count database mirror used in [8] — is both-sides unit; its faces are 36 unit squares and 12 dP_6 hexagons, so its hypersurface has 48 singular points, all of type (S), while its polar, the 12-vertex free sum $H^\circ \oplus H^\circ$, has 72 smooth triangular faces and smooth hypersurface.

Example 5.4 (the unique smooth mirror pair)**.** Exactly eight of the 590 polytopes have every two-dimensional face smooth, so that X itself is smooth. For exactly one of them is the polar of the same kind: the 24-cell — self-dual, with 24 octahedral facets, 96 unimodular triangular faces, and no lattice points besides the origin and its 24 vertices. Its generic anticanonical hypersurface is the classical self-mirror Calabi–Yau threefold with $(h^{1,1}, h^{2,1}) = (20, 20)$ and $\chi = 0$; the $(1,1)$-threefolds of Braun [3] are free quotients of special symmetric members of this family. The scan adds the uniqueness: among all 473,800,776 reflexive 4-polytopes, the 24-cell is the only one whose generic anticanonical hypersurface and mirror are *both* smooth.

6. The unique frozen mirror pair

Theorem 6.1. *Let*

$$\begin{gathered}\Delta_{\mathbb{F}_1} = \operatorname{conv}\{(1,0,0,0),\ (0,1,0,0),\ (1,-1,0,0),\ (0,0,1,0),\ (0,0,0,1),\ (0,0,1,-1),\\ (0,0,-1,1),\ (0,0,0,-1),\ (0,0,-1,0),\ (-1,1,0,0),\ (0,-1,0,0),\ (-1,1,-1,1),\\ (0,-1,0,-1),\ (0,-1,-1,0),\ (-1,1,-1,0),\ (-1,0,0,-1),\ (-1,0,-1,1),\\ (-1,-1,0,-1),\ (-2,1,-1,1),\ (-2,1,-1,0),\ (-1,-1,-1,0),\ (-2,0,-1,0)\} \subset \mathbb{Z}^4.\end{gathered}$$

Then $\Delta_{\mathbb{F}_1}$ is a both-sides unit reflexive polytope with 22 vertices and 26 facets, and:

(1) *its 72 two-dimensional faces are 50 smooth triangles, 20 unit squares, one dP_6 hexagon, and one quadrilateral $GL_2(\mathbb{Z})$-equivalent to the $\mathbb{F}_1$ polygon; every dual edge has lattice length one;*

(2) *the generic anticanonical $X \subset \mathbb{P}_{\Delta_{\mathbb{F}_1}}$ is a Calabi–Yau threefold whose singular locus consists of 20 ordinary double points, one dP_6-cone point, and one point with germ the anticanonical cone over $\mathbb{F}_1$; by* [8, Cor. 4.3]*, X admits no smoothing;*

(3) *the polar $\Delta^\circ_{\mathbb{F}_1}$ (26 vertices, 22 facets) is both-sides unit with 68 two-dimensional faces: 38 smooth triangles, 26 unit squares, two dP_7 pentagons and two dP_6 hexagons; the mirror X° has 26 ordinary double points, two dP_7-cone points and two dP_6-cone points, each with a local smoothing;*

(4) *the maximal projective crepant partial resolutions satisfy* $(h^{1,1}, h^{2,1})(\widehat{X}) = (20, 26)$ *and* $(h^{1,1}, h^{2,1})(\widehat{X^\circ}) = (26, 20)$*;*
(5) $\Delta_{\mathbb{F}_1}$ *is the* only *both-sides unit reflexive* 4*-polytope, up to lattice isomorphism, carrying a non-smoothable face that is not a triangle.*

Proof. (1)–(4) are finite verifications in exact arithmetic, performed by the ancillary file `both_sides_census.py`: the $\mathbb{F}_1$ identification is an explicit $GL_2(\mathbb{Z})$-equivalence of edge multisets, and the Hodge numbers are computed by the implementation of [8], anchored there on the quintic. The mirror swap in (4) is forced by Batyrev duality and serves as a consistency check. (5) is Theorem 1.4: the scan of the complete classification finds no other non-triangle non-smoothable face on any both-sides unit polytope. □

Remark 6.2 (a frozen threefold with a mobile mirror). The pair of Theorem 6.1 pairs a Calabi–Yau threefold that cannot be smoothed — its $\mathbb{F}_1$-cone point is rigid, with reduced miniversal base a point, the local model of Gross's exceptional case — with a mirror all of whose singular points move locally. Whether X° moves *globally* is exactly the frontier of [9]: all $26 + 2 + 2$ of its singular points sit over faces with $\ell(F^\circ) = 1$, so the single-face smoothing techniques of [9] do not apply; the 26 nodes require a cross-face Friedman rescue [9, §7]; and the del Pezzo cone points are the $\ell = 1$ germs that [9, Prop. 6.1, Rem. 6.2] place beyond Gross's theorems, with no Friedman-type obstruction known. The pair thus concentrates, on two mirror-dual threefolds, every open smoothability phenomenon of this series.

7. Questions

Question 7.1. Why triangles, zonotopes, and reflexive polygons? Theorem 1.3 is an exhaustive fact about dimension four; a conceptual, dimension-free proof — a local characterization of the unit-edge polygons admitting an all-unit, all-transversally-smooth reflexive completion — would explain it. The failed local balancing statement of §5.3 shows the answer is subtler than the star of the face; the appearance of exactly the *reflexive* polygons as the non-zonotope exceptions is suggestive in a polar-duality-symmetric problem.

Question 7.2. Is the mirror X° of Theorem 6.1 smoothable? All its germs are of type (S) with $\ell(F^\circ) = 1$. A positive answer would exhibit a Batyrev mirror pair with one member rigid against smoothing and the other smoothable — a maximally asymmetric pair for the deformation–resolution exchange of mirror symmetry; a negative answer for the nodes alone would already make X° a 26-node analogue of the one-node obstructed threefold of [9, Thm. 5.1].

Question 7.3. Corti–Filip–Petracci relate smoothing components of $C(Q)$ to 0-mutable Laurent polynomials with Newton polygon Q [4], and Filip [5] treats the non-isolated case. In that language, what distinguishes the twelve face classes of Table 2 — and what is the mirror-theoretic meaning of the unique $\mathbb{F}_1$ face? One would like to read Theorem 1.4 off the canonical scattering diagram of the mirror.

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